

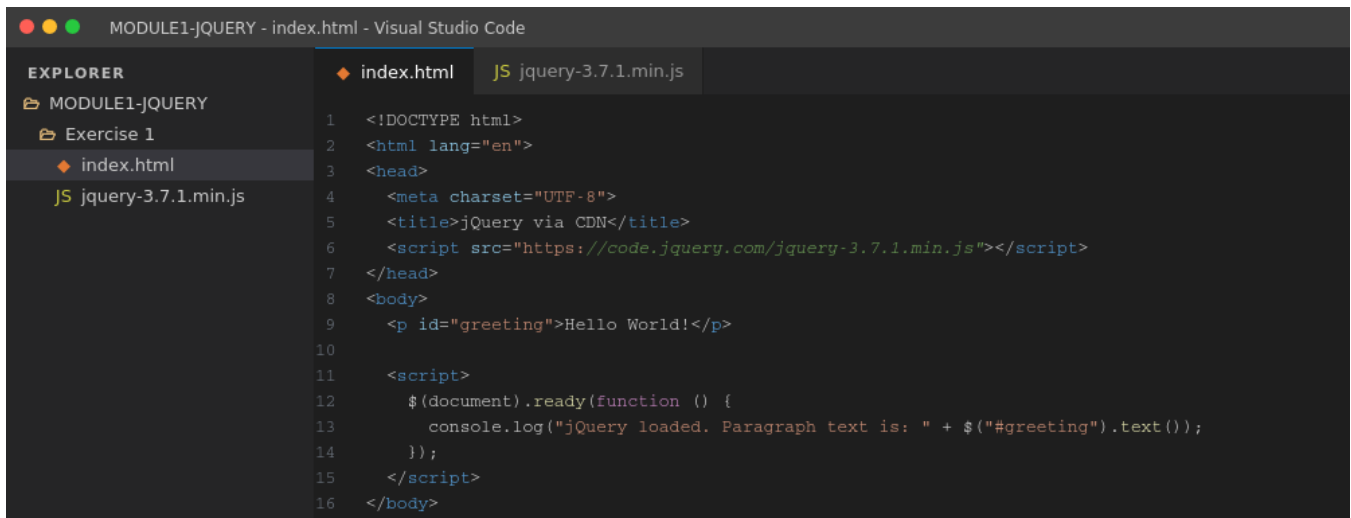
Module 1-jQuery

1. jQuery Library Inclusion

Exercise 1: Adding jQuery to a Page

- **Task:** Create a simple HTML page and include jQuery in two ways:
 1. Using a CDN link.
 2. Using a locally downloaded copy.
- **Bonus:** Add a `<p>` element that says "Hello World!" and log it using jQuery to the console.

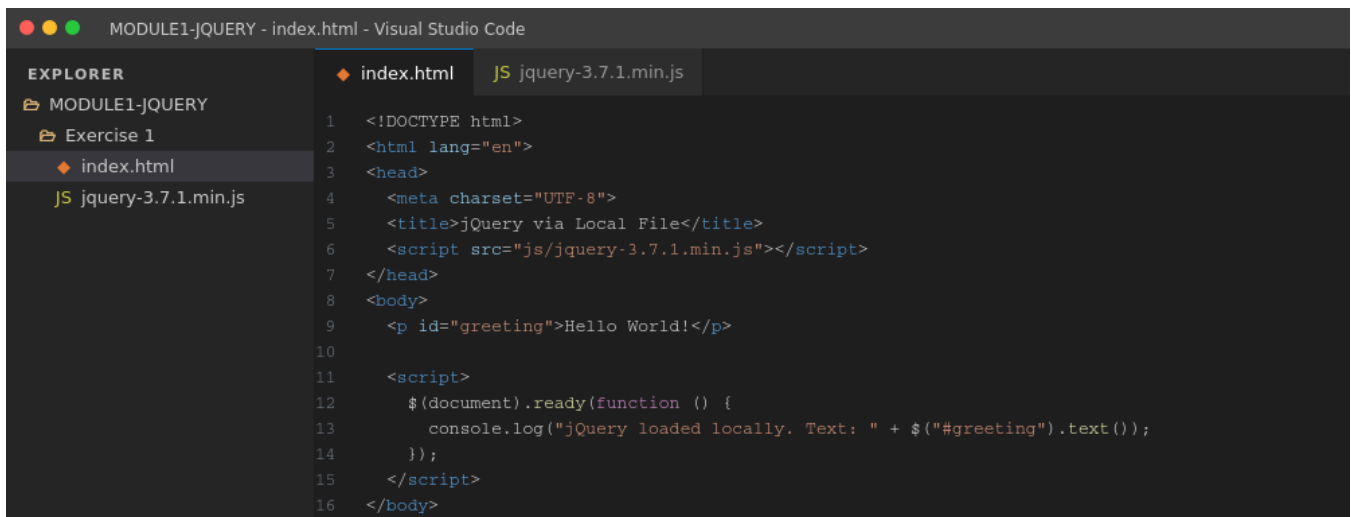
Version A — Using a CDN link (jquery_cdn.html)



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>jQuery via CDN</title>
6   <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7 </head>
8 <body>
9   <p id="greeting">Hello World!</p>
10
11   <script>
12     $(document).ready(function () {
13       console.log("jQuery loaded. Paragraph text is: " + $("#greeting").text());
14     });
15   </script>
16 </body>
```

Version B — Using a locally downloaded copy (jquery_local.html)

Download jquery-3.7.1.min.js from jquery.com and save it in the same folder as the HTML file, then reference it with a relative path instead of a URL.



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>jQuery via Local File</title>
6   <script src="js/jquery-3.7.1.min.js"></script>
7 </head>
8 <body>
9   <p id="greeting">Hello World!</p>
10
11   <script>
12     $(document).ready(function () {
13       console.log("jQuery loaded locally. Text: " + $("#greeting").text());
14     });
15   </script>
16 </body>
```

Output

Hello World!

Browser output — console prints: "jQuery loaded. Paragraph text is: Hello World!"

2. The \$() Function

Exercise 2: Selecting Elements

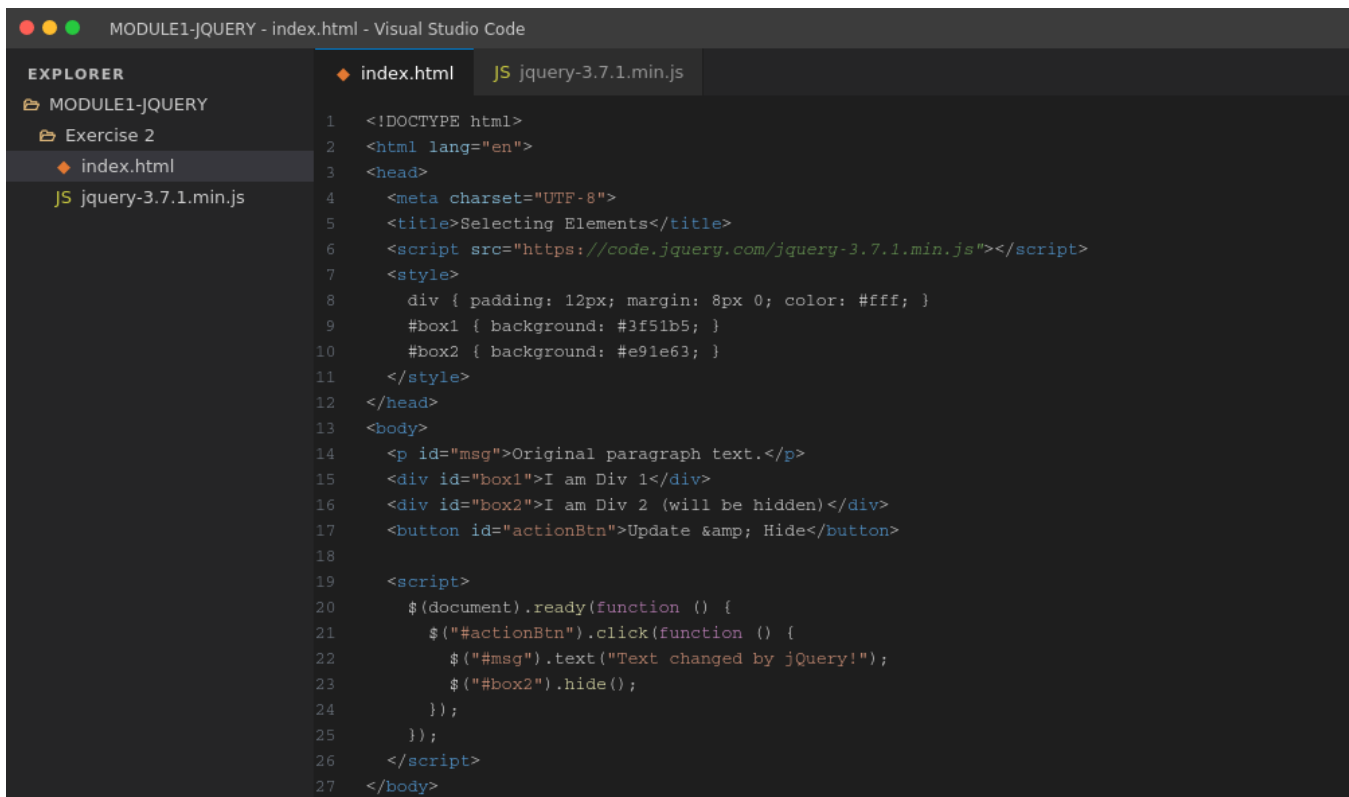
• Task:

Create a webpage with: 1 <p>, 2 <div>, and 1 <button>.

Use \$(document).ready() to ensure the DOM is loaded.

Use the \$() function to:

- Change the text of the <p>
- Hide one of the <div> elements when the button is clicked



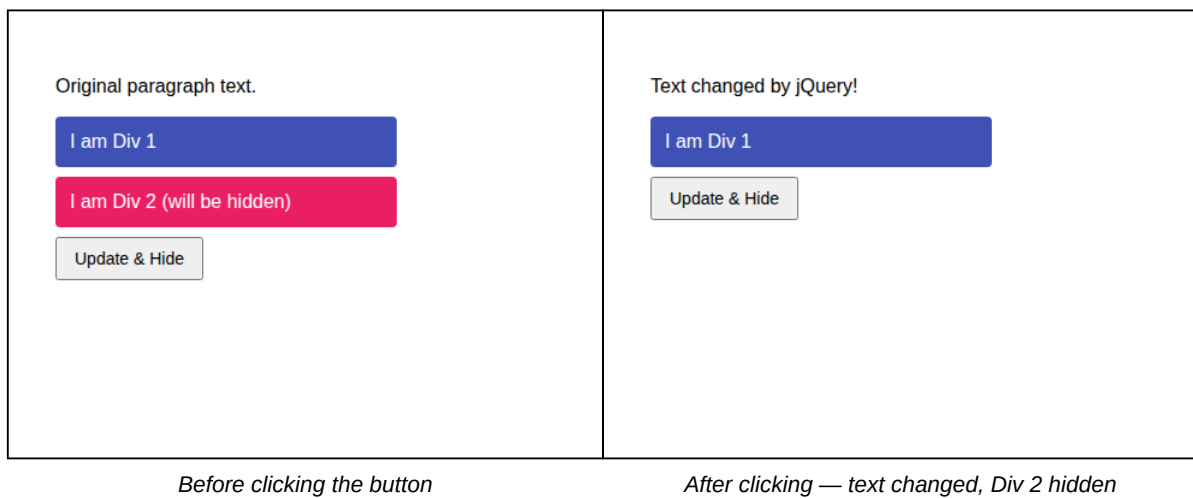
```
MODULE1-JQUERY - index.html - Visual Studio Code

EXPLORER
  MODULE1-JQUERY
    Exercise 2
      index.html
      jquery-3.7.1.min.js

index.html  JS jquery-3.7.1.min.js

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <title>Selecting Elements</title>
6    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7    <style>
8      div { padding: 12px; margin: 8px 0; color: #fff; }
9      #box1 { background: #3f51b5; }
10     #box2 { background: #e91e63; }
11   </style>
12 </head>
13 <body>
14   <p id="msg">Original paragraph text.</p>
15   <div id="box1">I am Div 1</div>
16   <div id="box2">I am Div 2 (will be hidden)</div>
17   <button id="actionBtn">Update & Hide</button>
18
19   <script>
20     $(document).ready(function () {
21       $("#actionBtn").click(function () {
22         $("#msg").text("Text changed by jQuery!");
23         $("#box2").hide();
24       });
25     });
26   </script>
27 </body>
```

Output



3. jQuery Methods

Exercise 3: Use of Common Methods

- **Task:**

Add 3 `<div>` elements with class `.box`

Use the following jQuery methods: `.hide()`, `.show()`, `.fadeOut()`, `.fadeIn()`, `.toggle()`

Apply these methods when clicking respective buttons.

- **Bonus:** Chain methods like `.slideUp().delay(1000).slideDown()`

```
MODULE1-JQUERY - index.html - Visual Studio Code

EXPLORER
  MODULE1-JQUERY
    Exercise 3
      index.html
      JS jquery-3.7.1.min.js

index.html  JS jquery-3.7.1.min.js

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <title>Common Methods</title>
6    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7    <style>
8      .box {
9        width: 150px; height: 60px; margin: 6px 0;
10       background: #ff9800; color: #fff;
11       display: flex; align-items: center; justify-content: center;
12     }
13   </style>
14 </head>
15 <body>
16   <div class="box">Box 1</div>
17   <div class="box">Box 2</div>
18   <div class="box">Box 3</div>
19
20   <button id="hideBtn">Hide</button>
21   <button id="showBtn">Show</button>
22   <button id="fadeOutBtn">Fade Out</button>
23   <button id="fadeInBtn">Fade In</button>
24   <button id="toggleBtn">Toggle</button>
25   <button id="chainBtn">Slide Up, Wait, Slide Down</button>
26
27   <script>
28     $(document).ready(function () {
29       $("#hideBtn").click(function () { $(".box").hide(); });
30       $("#showBtn").click(function () { $(".box").show(); });
31       $("#fadeOutBtn").click(function () { $(".box").fadeOut(); });
32       $("#fadeInBtn").click(function () { $(".box").fadeIn(); });
33       $("#toggleBtn").click(function () { $(".box").toggle(); });
34
35       class="cmt">// Bonus: chained methods
36       $("#chainBtn").click(function () {
37         $(".box").slideUp(400).delay(1000).slideDown(400);
38       });
39     });
40   </script>
41 </body>
42 </html>
```

Output



Before — all 3 boxes visible

After clicking Fade Out

4. DOM Manipulation

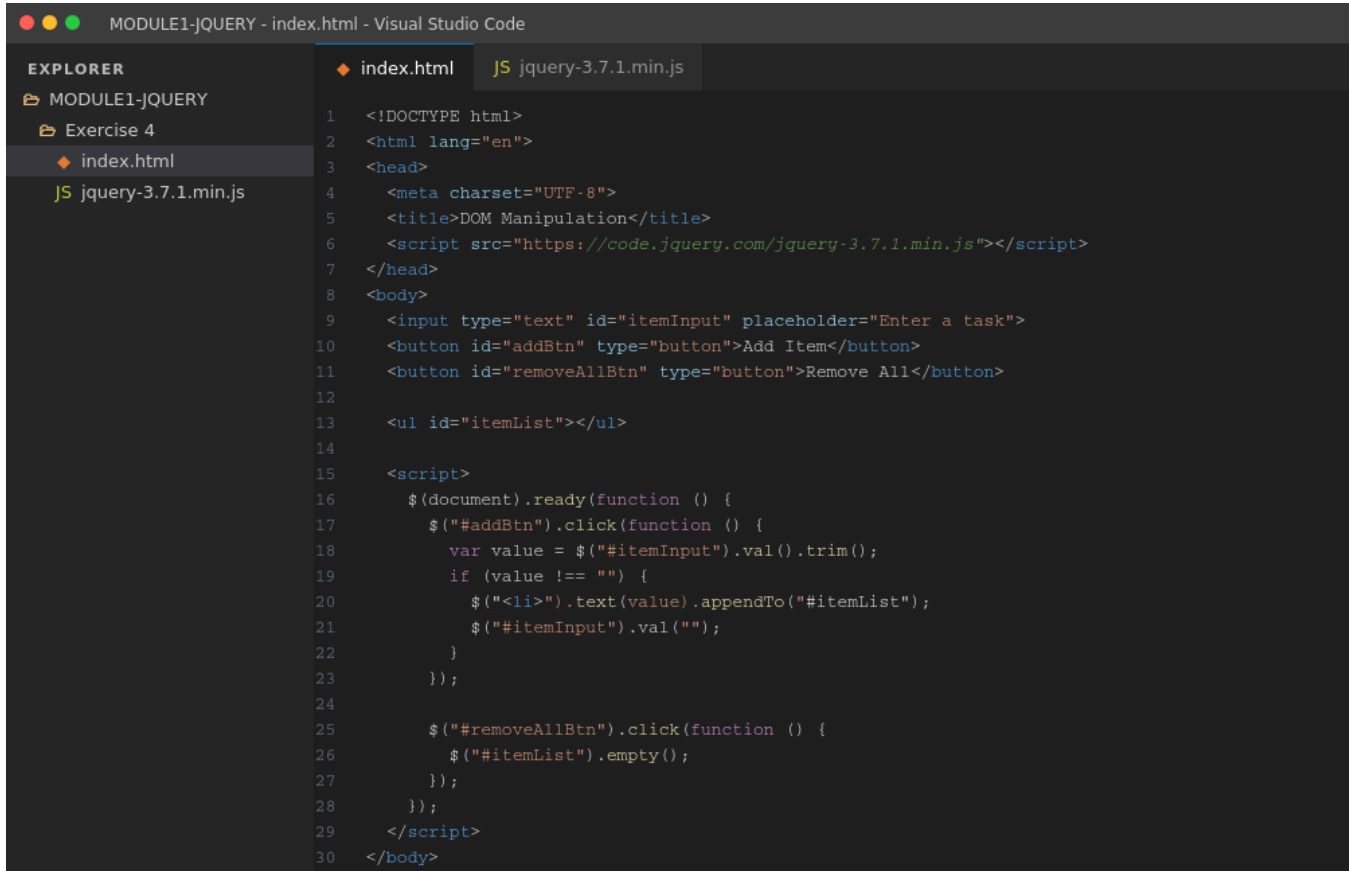
Exercise 4: Creating and Modifying Elements

• Task:

Create a form with an input field and a submit button.

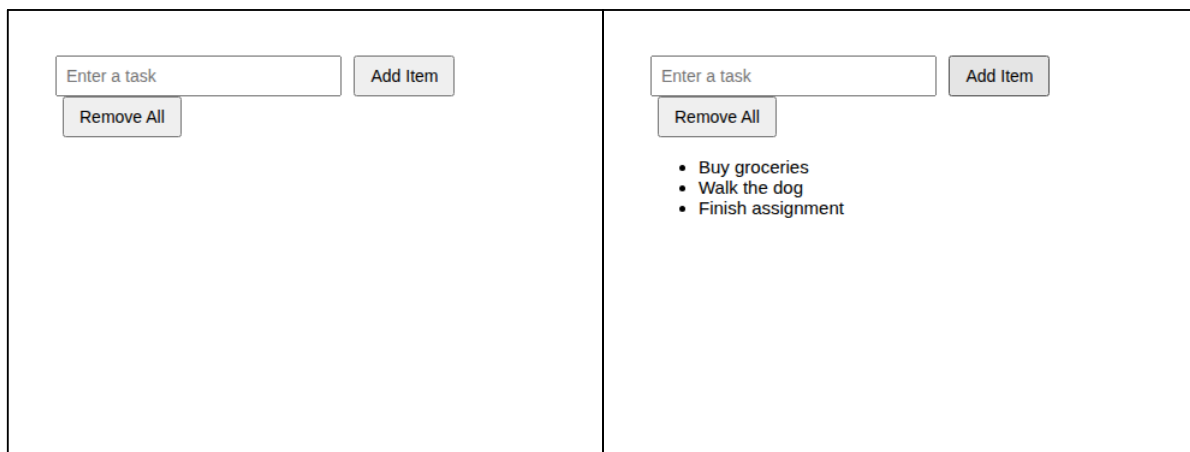
When the button is clicked: take the value from the input, create a new `<li>` with that value, and append it to a `<ul>`.

Add a "Remove All" button that empties the `<ul>`.



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>DOM Manipulation</title>
6   <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7 </head>
8 <body>
9   <input type="text" id="itemInput" placeholder="Enter a task">
10  <button id="addBtn" type="button">Add Item</button>
11  <button id="removeAllBtn" type="button">Remove All</button>
12
13  <ul id="itemList"></ul>
14
15  <script>
16    $(document).ready(function () {
17      $("#addBtn").click(function () {
18        var value = $("#itemInput").val().trim();
19        if (value !== "") {
20          $("#<li>").text(value).appendTo("#itemList");
21          $("#itemInput").val("");
22        }
23      });
24
25      $("#removeAllBtn").click(function () {
26        $("#itemList").empty();
27      });
28    });
29  </script>
30 </body>
```

Output



Before — empty list

After adding 3 items

5. Working with Events

Exercise 5: Handling User Interactions

- **Task:** Create a page with:

A <button> with id #colorBtn

A <div> with id #colorBox

Use jQuery to:

Change the background color of #colorBox to red when the button is clicked.

Change it back to white when double-clicked.

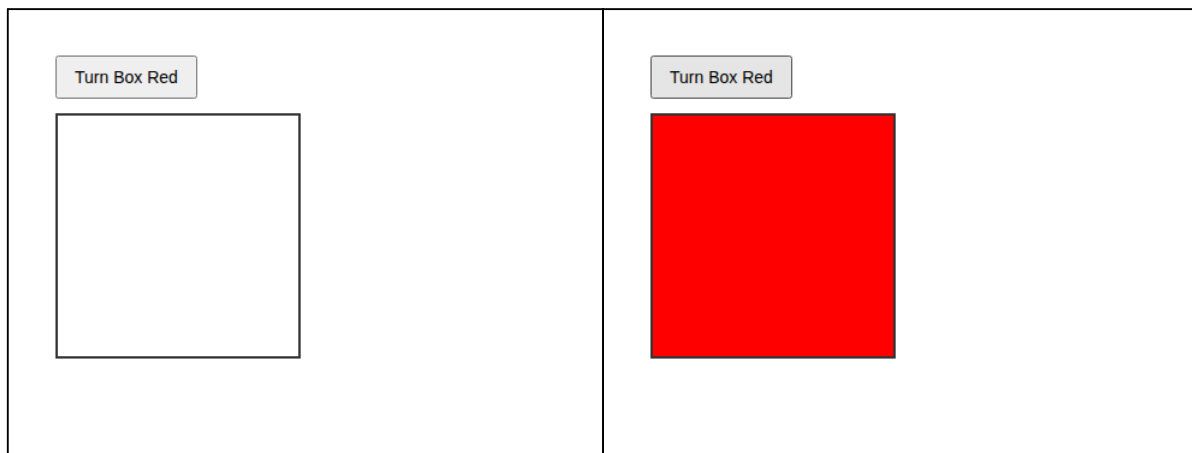
```
MODULE1-JQUERY - index.html - Visual Studio Code

EXPLORER
  MODULE1-JQUERY
    Exercise 5
      index.html
      JS jquery-3.7.1.min.js

index.html  JS jquery-3.7.1.min.js

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <title>User Interactions</title>
6    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7    <style>
8      #colorBox {
9        width: 200px; height: 200px; margin-top: 12px;
10       background: white; border: 2px solid #333;
11     }
12   </style>
13 </head>
14 <body>
15   <button id="colorBtn">Turn Box Red</button>
16   <div id="colorBox"></div>
17
18   <script>
19     $(document).ready(function () {
20       $("#colorBtn").click(function () {
21         $("#colorBox").css("background-color", "red");
22       });
23
24       $("#colorBox").dblclick(function () {
25         $(this).css("background-color", "white");
26       });
27     });
28   </script>
29 </body>
```

Output



Before — white box

After clicking Turn Box Red

6. Event Helpers

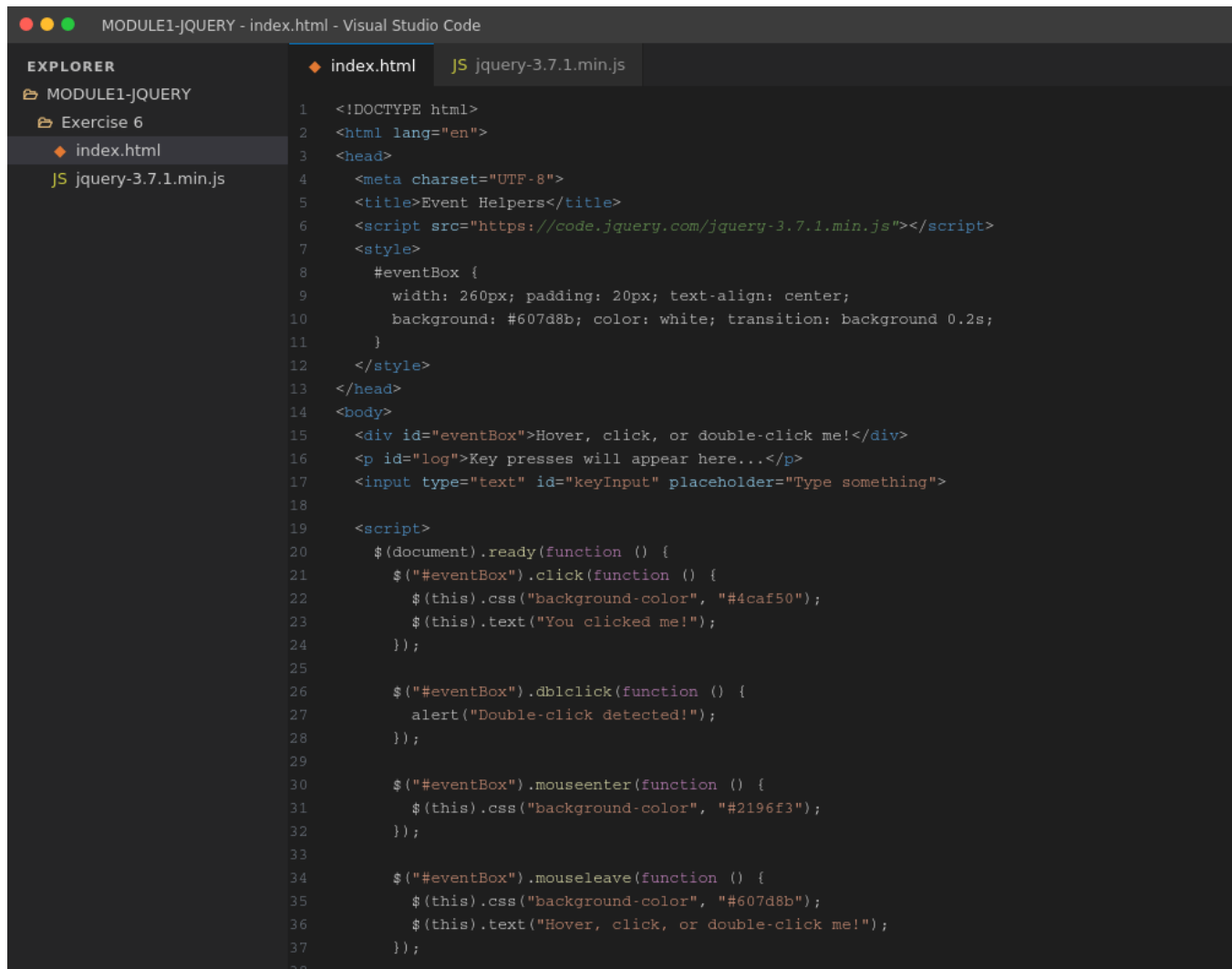
Exercise 6: Mouse and Keyboard Events

• Task:

Create a <div> with some text.

Use the following event helpers: .click(), .dblclick(), .mouseenter() and .mouseleave(), .keypress() on an input field.

Make each event do something visible (e.g., change background color, add text, show an alert, etc.).



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Event Helpers</title>
6   <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7   <style>
8     #eventBox {
9       width: 260px; padding: 20px; text-align: center;
10      background: #607d8b; color: white; transition: background 0.2s;
11    }
12  </style>
13 </head>
14 <body>
15   <div id="eventBox">Hover, click, or double-click me!</div>
16   <p id="log">Key presses will appear here...</p>
17   <input type="text" id="keyInput" placeholder="Type something">
18
19   <script>
20     $(document).ready(function () {
21       $("#eventBox").click(function () {
22         $(this).css("background-color", "#4caf50");
23         $(this).text("You clicked me!");
24       });
25
26       $("#eventBox").dblclick(function () {
27         alert("Double-click detected!");
28       });
29
30       $("#eventBox").mouseenter(function () {
31         $(this).css("background-color", "#2196f3");
32       });
33
34       $("#eventBox").mouseleave(function () {
35         $(this).css("background-color", "#607d8b");
36         $(this).text("Hover, click, or double-click me!");
37       });
38     });
39   </script>
40 </body>
41 </html>
```

Output

Hover, click, or double-click me! Key presses will appear here... Type something	You clicked me! Key presses will appear here... Type something
---	---

Default state

After clicking the box

You clicked me!

Last key pressed: H

H

After typing "H" in the input — status line updates live